

Analysis of genetic diversity and population structure in some Moroccan alfalfa populations (*Medicago sativa* L.) based on SSR markers

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ABSTRACT

Genetic diversity assessments represent the foundation and a prerequisite for any well-considered conservation, exploitation, or further investigation of indigenous genetic resources. Indeed, information regarding the genetic diversity and population structure of Moroccan alfalfa populations can be invaluable for the effective utilization of these genetic resources. In our study, a total of 280 individuals of *Medicago sativa* L. from eight populations (six Moroccan populations and two introduced populations) were genotyped using seven SSR markers to elucidate their genetic diversity and population structure. These seven markers exhibited high levels of polymorphism (PIC=0.848), resulting in a total of 137 alleles being identified. The present study unveiled substantial levels of genetic diversity within the analyzed alfalfa populations ($A=7.464$, $A_i=2.281$, $HS=0.648$, $HE=0.640$, $I=0.129$, and $PPL=36.95\%$). Analysis of AMOVA revealed that 16%, 27%, and 56% of the total genetic variation occurred among regions, among populations, and within populations, respectively. The substantial variation within populations was anticipated due to the allogamy and autotetraploidy of alfalfa. The results of UPGMA cluster analysis and PCoA demonstrated that the Moroccan populations could be grouped into two main clusters, which were significantly associated with their geographical origins. These findings substantiate the dual genetic origin of Moroccan alfalfa populations.

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1. Introduction:

Alfalfa (*Medicago sativa* L.) is one of the most important forage crops in the world [1]. Alfalfa is one of world's most valuable fodder crops, planted for hay and pasture in more than 80 countries with an area of 30-35 million ha [2-3]. The main alfalfa-producing areas are central and eastern Europe, the Mediterranean basin, and North America.

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In Morocco, alfalfa (*Medicago sativa* L.) is the main forage legume, it occupies more than 106 000 ha, i.e. approximately 22,8% of the total area devoted to fodder [4]. Its use for centuries in many areas and its crop diversity has enabled the development of landraces adapted as those in the traditional oasis (Erfoud and Rich) and mountains agro-ecosystems regions (Demnate) [5] with agronomic traits of interest [6-7]. Therefore, these Moroccan alfalfa populations, which have been in the hands of regional farmers for many years and adapted to the region, appear to be an important source for breeding programs. However, their genetic diversity is increasingly threatened by various factors of genetic erosion such as the introduction of improved varieties, prolonged drought, diseases, and unavailability of seeds [8]. In order to make sufficient use of these landraces, it is necessary to know their genetic properties. Indeed, the study of the genetic diversity of these local populations of alfalfa is a primordial step that must precede any reasoned operation of conservation, exploitation or further study of indigenous genetic resources.

Genetic diversity refers to the variation in the range of different inherited traits or macromolecules within a population or a species. Genetic diversity is viewed as an important aspect to understand the dynamics of plant populations and a reduction in genetic diversity may limit the ability of the populations to evolve and persist in nature [9]. Indeed, most researchers agree that genetic diversity is crucial because it helps to maintain the population's adaptability in the face of diseases, pests, climate change, and other stressors [10]. A variety of approaches have been used to characterize genetic diversity in alfalfa including phenotypic markers, isozymes [5], and molecular markers consisting of AFLPs [11], RAPDs [12], RFLPs [13], mitochondrial DNA [14], and simple sequence repeat markers SSRs [15]. Phenotypic markers and protein markers are limited in number, affected by the environment, and represent only a part of the genetic diversity [16]. Compared with these DNA-based markers, SSRs have been used extensively for the development of new varieties studies, and has a range of advantages, they are highly reproducible, co-dominantly inherited, highly polymorphic, abundant markers in genomes, occur frequently in plants, and can function with low-quality DNA [17-19]. It has been shown that SSR markers can be used in plants for genetic mapping, genetic diversity assessment, population genetics, and marker-assisted selection [20]. They have provided species-specific allele patterns in alfalfa and can be useful markers for determining the genetic variation and the differences between plant varieties [21]. SSR markers have been first developed in *Medicago sativa* by Diwan et al., (1997) [22]. Other SSRs originating from *M. truncatula* have since also been used to amplify and reveal polymorphism in alfalfa [23; 24]. Since then, a growing number of studies have used SSR markers for estimating the genetic diversity in alfalfa.

No attempts have been made so far to characterize the widely used Moroccan alfalfa populations using SSR markers. Because of allogamy, tetraploidy of alfalfa, and its ancient history in Morocco, a high level of genetic diversity may be expected among individual plants as well as within and between populations.

Thus, the objectives of the present study were to (i) identify polymorphic SSR markers with high informative value for the Moroccan alfalfa in terms of population differentiation and identification, (ii) estimate the level of genetic diversity present in Moroccan alfalfa populations, and (iii) determinate the degree of genetic relationship between landraces obtained from different geographical regions.

2. Materiel and Methods:

2.1. Plant materiel:

Six Moroccan alfalfa populations, *Medicago sativa* L., were used in this study: two populations Demnate19 and Demnate21 that are originally from Demnate (31° 43' 52" nord, 7° 02' 10" ouest), two populations Rich7 and Rich8 from region of Rich (32° 16' 22" nord, 4° 30' 26" ouest), and two populations Erfoud5 and Erfoud6 originally in the Erfoud region (31° 26' 20" nord, 4° 14' 37" ouest) (Figure 1). We also analyzed two introduced populations, Siriver and Tamantit, shown in Table1. This study was carried out in experiment field of Tassaout (INRA Marrakech). Thirty-five plants per population were randomly selected, and green healthy leaves from each plant were collected for DNA extraction.

Table 1. List of various studied populations of alfalfa.

Populations		Geographical medium	Locality and Origin
Local populations	Rich7, Rich8	Oasis area of desert valley of Ziz	High Ziz; Rich; Morocco
	Erfoud5, Erfoud6	Oasis area of desert valley of Ziz	Plain area of Ziz; Erfoud; Morocco
	Demnate19, Demnate21	High Atlas mountain	Demnate; Morocco
Introduced populations	Tamantit		Algeria
	Siriver		Italy

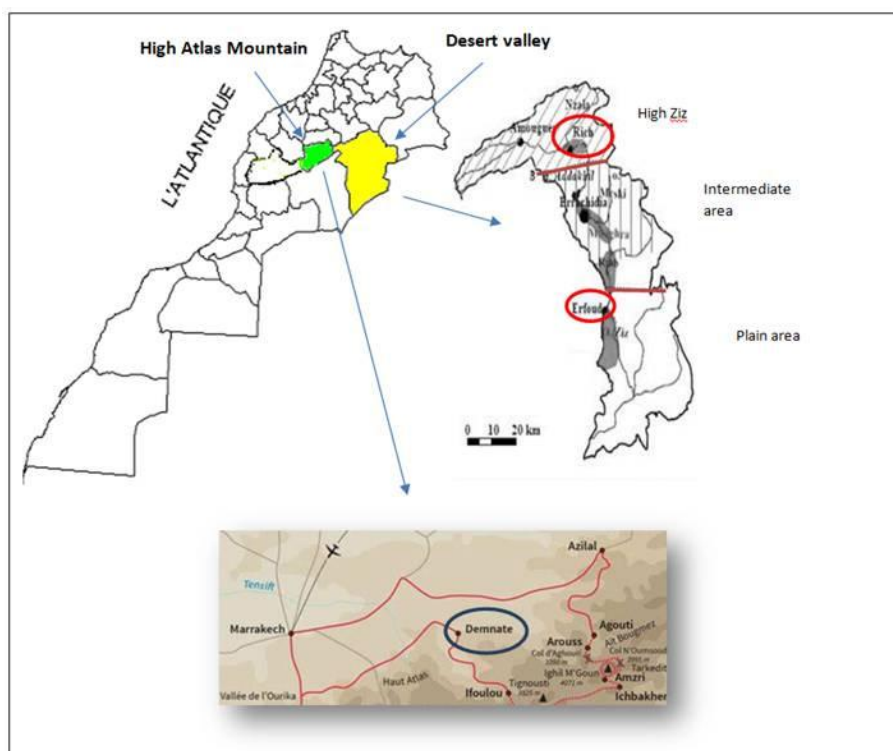


Figure 1. The geographical distribution of Moroccan alfalfa populations.

2.2 DNA extraction:

Genomic DNA was extracted from young leaves of each plant using a Genomic DNA Purification Kit (WIZARD) according to the manufacturer's protocol (Promega) and stored at -20°C until needed. DNA was quantified spectrophotometrically and diluted to $40\text{ng}/\mu\text{l}$ in double-distilled sterile water.

2.3 SSR amplification:

Seven primer pairs detecting SSR loci in alfalfa as described by Diwan *et al.* (1997) [22] and Julier *et al.*, (2003) [24], were used in the amplification reactions (Table 2). PCR was performed in a total volume of $20\ \mu\text{l}$ containing $37.5\ \text{ng}$ of template DNA, $0.2\ \text{mM}$ of each deoxynucleotide, $1.5\ \text{mM}$ MgCl_2 , $0.4\ \mu\text{M}$ of each primer, $1\times$ GoTaq Flexi Buffer and 0.5 unit of GoTaq Flexi DNA polymerase (Promega). An initial GoTaq polymerase activation step of $4\ \text{min}$ at 94°C , followed by a step of 35 cycles with $30\ \text{secs}$ at 94°C , $1\ \text{min}$ at 55°C and $1\ \text{min}$ at 72°C , plus a final elongation step of $7\ \text{min}$ at 72°C were performed. PCR products ($6\ \mu\text{L}/\text{lane}$) were separated by electrophoresis for $2\ \text{h} - 3\ \text{h}$ under constant voltage ($5\text{v}/\text{cm}$) in 4% SFR (Small Fragment Resolution AMRESCO) agarose gel containing BET ($4.16\ \mu\text{g}/\text{ml}$), and submersed in $1\times$ TBE buffer. Gels were photographed. The PCR marker (Promega) was also run on each gel as a molecular weight standard.

2.4 SSR analysis:

The genotypes of 280 individuals were detected by 7 pairs of SSR markers which were distributed on 7 chromosomes of alfalfa (Table 2). The different bands of genotypes were defined according to their base weight. For each individual the number of different alleles was determined. Allele dose were scored. The raw gel data were scored as allele sizes for further formatting as input files for statistical analysis. The genetic data were analyzed using multiple software combinations. AUTOTET [25] was used to calculate for each population the allelic richness (A), allelic richness within individuals (A_i), genotypic richness (G), observed heterozygosity (H_o), and expected heterozygosity (H_e). The same software was also used to calculate the number of observed alleles (N_a), and effective number of alleles (N_e) for each SSR loci. DISPAN program [26] was used to estimate for each locus the genetic diversity (HS). The polymorphism information content (PIC) of SSR markers was calculated using the formula $PIC = 1 - \sum(P_{ij})^2$, where P_{ij} is the frequency of the i th allele in the j th population, for each SSR locus according to Botstein *et al.*, (1980) [27]. SPAGeDi version $1.5d$ [28] was used to calculate the inbreeding coefficient (F_i) (for each population and for each locus), gene diversity H_s (for each population), standard genetic distance (D_s) Nei's 1978 [29] and the population genetic differentiation coefficient (F_{ST}) (for each locus). GenAlEx version 6.503 [30] was used to calculate the shannon information index (I) and percentage of polymorphic loci (PPL). In addition, an analysis of molecular variance (AMOVA) was conducted using GenAlEx version 6.503 to calculate the sum of squares and variance components within populations and between populations. Statistical significance was evaluated using 1000 substitutions. Further analysis of genetic structure was done by Principal Coordinate Analysis (PCoA) using GenAlEx 6.503 .

[30]. Cluster analysis was performed based on Nei's standard genetic distance [31] by means of UPGMA (unweighted pair - group method using arithmetic average) clustering procedure, using POPTREEW [32]. The frequencies of null alleles (r) were estimated from the heterozygote deficiency following Brookfield (1996) [33] as $r = (H_E - H_O)/(1 + H_E)$ for each locus and each population.

3. Results:

3.1. SSR polymorphism and Genetic diversity:

The gels obtained with each of the seven microsatellite loci were read taking into account the allele band number, base pair weight, and allele dose. The abovementioned seven SSR markers amplified a total of 137 alleles in a set of eight cultivated alfalfa populations. The number of observed alleles per SSR locus (N_a) ranged from 10 (for AFca16) to 35 (for B14B03) (Table2). The majority of the loci scored in this study were highly polymorphic, having 19 alleles per locus on average. The lowest and highest base pair weight ranged from 80bp for AFca16 primer to 250bp for AFct11 primer (Table2).

Table 2. Population-genetic data for seven SSR loci in alfalfa populations.

Loci	Core motif	Number of alleles (N_a)	Range of allele size (pb)	Average number of alleles per plant (A_i)	Number of different alleles per plant (G)				Linkage group
					4	3	2	1	
AFca1	(CT) ⁴ (CA) ¹⁰	15	138 – 206	2.57 (1-4)	0.13	0.48	0.21	0.16	4
AFca11	(CA) ¹¹	16	131 – 190	1.82 (1-3)	0.01	0.08	0.65	0.27	6
AFca16	(CA) ¹²	10	80 – 116	1.17 (1-2)	0.00	0.00	0.31	0.69	8
AFct11	(CT) ¹²	20	174 – 250	2.44 (1-4)	0.08	0.39	0.43	0.07	7
AFct32	(CT) ¹⁴	28	98 – 196	2.66 (1-4)	0.20	0.38	0.33	0.07	3
B14B03	(CA) ⁹	35	100 – 223	2.44 (1-4)	0.11	0.37	0.38	0.13	5
MTLEC2A	(AT) ¹⁹	12	118 – 234	2.51 (1-4)	0.16	0.37	0.32	0.12	3
Average	-	19.43	-	2.23	0.10	0.30	0.38	0.22	-

The average number of alleles observed per plant at each locus (A_i) ranged from 1.17 in AFca16 to 2.66 in AFct32 with an average of 2.23 for all loci (Table2). The proportion of plants containing one allele, two, three and four different alleles at each locus is given in Table2. The frequency of plants with either one allele or four different alleles was less than those with two and three alleles in a population. B14B03 was the most informative locus for population identification, with one specific 149 pb allele being present in all genotypes of Rich8 population and was not detected in other populations. MTLEC2A was also informative for population identification, with one specific 124 pb allele being present in most of genotypes of Tamantit cultivar.

All SSR loci had high polymorphism levels, with PIC values ranging from 0.747 (AFca11) to 0.936 (B14B03) and an average PIC value of 0.848. The average level of genetic diversity (H_s) was 0.643. The levels of expected heterozygosity (H_E) varied from 0.399 (AFca11) to 0.788 (AFct32), and the average level was 0.640. The values of observed heterozygosity (H_O) ranged from 0.167 (AFca16) to 0.699 (AFct32) with an average level of 0.552 over all loci (Table3). Among all loci, only AFca1, AFca11 and MTLEC2A had negative inbreeding coefficient (F_i) values of -0.054, -0.061 and -0.013 respectively. However, AFct11, AFct32, B14B03 had positive F_i values of 0.039, 0.123, 0.200 respectively, AFca16 showed the highest F_i value of 0.722 (Table3). The recurrence of the most frequent and infrequent alleles followed a similar pattern across all populations for each of the seven SSR loci. No null homozygote was observed in our data (i.e. a PCR product was obtained from all individuals), suggesting that putative null alleles occurred at very low frequencies. Indeed, null alleles were detected at low frequencies (r) ranging from 0.001 for AFca1 to 0.088 for B14B03 across all populations, except for AFca16 locus r was the highest with a value of 0.269.

Table 3. Summary of genetic statistics for each locus in the alfalfa populations.

Locus	N_a	N_e	H_O	H_E	F_i	H_S	F_{ST}	PIC	r
AFca1	15	6.21	0.652	0.612	-0.054	0.614	0.293	0.838	0.001
AFca11	16	3.96	0.426	0.399	-0.061	0.406	0.494	0.747	0.007
AFca16	10	5.27	0.165	0.596	0.722	0.608	0.272	0.809	0.269
AFct11	20	10.74	0.650	0.673	0.039	0.673	0.278	0.906	0.020
AFct32	28	10.2	0.699	0.788	0.123	0.786	0.133	0.901	0.049
B14B03	35	15.81	0.621	0.770	0.200	0.772	0.188	0.936	0.088
MTLEC2A	12	4.94	0.656	0.642	-0.013	0.645	0.211	0.797	0.010
Average	19.4	8.161	0.552	0.640	0.137	0.643	0.263	0.848	0.063

N_a : number of observed alleles; N_e : effective number of alleles; H_O : observed heterozygosity; H_E : expected heterozygosity; F_i : inbreeding coefficient; H_S : genetic diversity; F_{ST} : genetic differentiation coefficient; PIC: polymorphism information content; r : frequencies of null alleles.

Among the eight studied populations, Erfoud5 contained the highest allelic richness ($A = 9.57$), and Rich8 and Tamantit ($A = 6.571$) showed the lowest allelic richness. On average, 7.464 alleles are observed per population (Table 4). The gene diversity (H_s) detected for each Moroccan population over seven SSR loci ranged from 0.575 for Rich8 population to 0.723 for Erfoud5 population based on Nei's genetic diversity index, showed greater levels of within population variability. For cultivars the gene diversity (H_s) ranged from 0.638 for Tamantit cultivar to 0.642 for Siriver cultivar (Table 4). Inbreeding coefficient (F_i) ranged from 0.003 (Rich7) to 0.211 (Erfoud6) for Moroccan alfalfa populations, and from 0.164 (Siriver) to 0.177 (Tamantit) for cultivars, with an average F_i of 0.149. The average of gene diversity (H_s), Observed heterozygosity (H_o), expected heterozygosity (H_e), and inbreeding coefficient (F_i) of the eight populations were 0.648, 0.552, 0.640, and 0.149, respectively. Among the populations, Erfoud5 had statistically ($P < 0.001$) the highest level of genetic diversity ($H_s = 0.723$; $H_e = 0.705$), and the Rich8 population had the lowest level of genetic diversity ($H_s = 0.575$; $H_e = 0.558$). The Tamantit cultivar showed the highest frequencies of null alleles (r), with an average of 0.143 (Table 4). In Rich7 population, null alleles (r) occurred at the lowest frequencies of -0.001 on average. For the other populations frequencies of null alleles was low and ranged from 0.033 to 0.080.

Table 4. Genetic diversity parameters among the eight populations of *Medicago sativa* L.

Populations	A	Ai	G	HS	HO	HE	Fi	I	r	PPL
Erfoud5	9.571	2.500	16.429	0.723	0.617	0.705	0.150	0.151	0.058	47.79%
Erfoud6	7.143	2.217	15.429	0.686	0.540	0.682	0.200	0.131	0.080	36.76%
Rich7	7.714	2.594	13.571	0.651	0.648	0.647	0.003	0.132	-0.001	38.24%
Rich8	6.571	1.976	12.857	0.575	0.446	0.558	0.200	0.108	0.066	31.62%
Tamantit	6.571	2.245	14.000	0.638	0.527	0.632	0.177	0.125	0.143	31.62%
Siriver	8.429	2.241	15.857	0.642	0.540	0.643	0.164	0.136	0.060	41.91%
Demt19	7.000	2.327	13.143	0.633	0.576	0.621	0.098	0.121	0.033	33.82%
Demt21	6.714	2.151	14.429	0.633	0.524	0.631	0.189	0.127	0.069	33.82%
Average	7.464	2.281	14.464	0.648	0.552	0.640	0.149	0.129	0.063	36.95%

A: allelic richness; Ai: allelic richness within individuals; G: genotypic richness; HS: gene diversity average over seven loci; HO: observed heterozygosity; HE: expected heterozygosity; Fi: inbreeding coefficient; I: shannon's information index; r: frequencies of null alleles; PPL: percentage of polymorphic loci.

3.2. Genetic differentiation and structure:

3.2.1. Genetic differentiation:

The genetic differentiation of populations determined by the analysis of molecular variance (AMOVA) (Table 5) revealed that only 28% of the total variation occurred among populations; 56% of the total variation was explained by the differences among genotypes within populations and the remaining 16% of the genetic variability occurred among regions. All results showed a high level of significance ($P < 0.001$).

Table 5. AMOVA (Analyze of Molecular Variance) for populations of *Medicago sativa* L.

Source	df	Sum of Squares	Means of Squares	Estimated Variation	Percentage of variation %	p value
Among Regs	4	962.857	240.714	1.963	16%	0.001
Among Pops	3	400.029	133.343	3.607	28%	0.001
Within Pops	272	1927.886	7.088	7.088	56%	0.001
Total	279	3290.771	-	12.658	100%	-

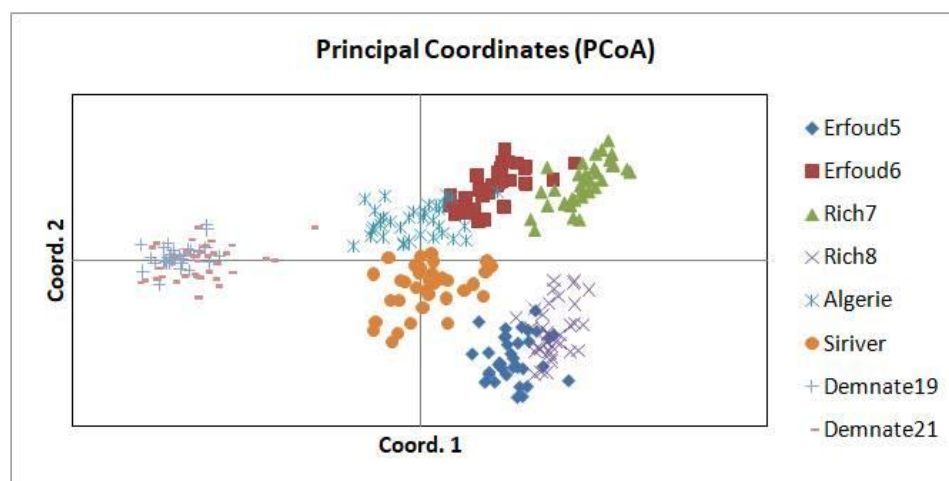
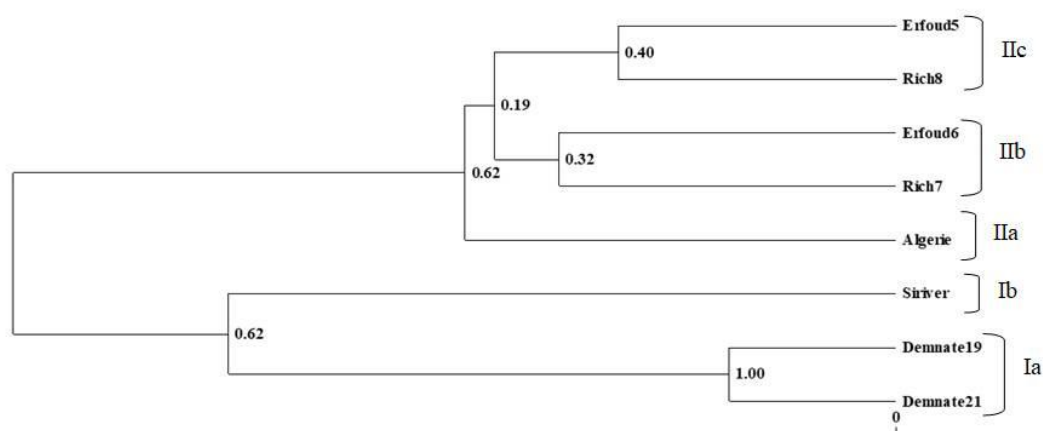
The genetic differentiation coefficient (F_{ST}) between the populations was 0.263 (Table 3). However, the value of F_{ST} alone does not give a precise idea of the genetic distance between populations. For this, and to better reveal the relationships between the 8 alfalfa genotypes, the quantity $F_{ST}/(1-F_{ST})$ (fixation genetic distance or wright's distance) was calculated according to ANOVA approach. This quantity measures the proportion of genetic variance that is due to differentiation between populations relative to the total genetic variance across all populations. The $F_{ST}/(1-F_{ST})$ values computed for each pairwise combination among the eight genotypes ranged from 0.1530 to 0.6428 (Table 6) and each of the pairwise values were significantly different ($P < 0.001$) (Table 6). The smallest fixation genetic distance (0.1530) was detected between the two populations (Demtate19 and Demtate21) originated from Demtate region. However, the highest fixation genetic distance (0.6428) was observed between the Demtate19 (Demtate region) population and the Rich8 population (Rich region). The two populations that originated from the Demtate region showed the highest genetic differentiation with the other populations (Rich and Erfoud). Meanwhile, the populations that originated from Rich and Erfoud regions showed smallest genetic differentiation between one another (Table 6).

Table 6. Fixation genetic distance among the 6 Moroccan alfalfa populations and the 2 cultivars.

Populations	Erfoud6	Rich7	Rich8	Tamantit	Siriver	Demt19	Demt21
Erfoud5	0.2004	0.2554	0.2393	0.1835	0.3691	0.3579	0.3468
Erfoud6	*****	0.2211	0.3606	0.2358	0.3019	0.3446	0.3541
Rich7		*****	0.2665	0.3157	0.3982	0.4914	0.4816
Rich8			*****	0.4518	0.5139	0.6428	0.6297
Tamantit				*****	0.4170	0.3649	0.3810
Siriver					*****	0.4034	0.3802
Demt19						*****	0.1530

3.2.2. Population structure:

The principal coordinate analysis (PCoA) was conducted to assess the population subdivisions (Figure 2). The first three principal coordinates explained 16.42%, 25.27% and 33.63%, respectively, and 75.32% of the total variation. The results of PCoA of the 280 alfalfa genotypes showed a clear separation of the Demnate populations from the Erfoud and Rich populations, and the intermediate position of the Siriver and Tamantit cultivars (Figure 2). Individuals in the Demnate19 population and Demnate21 population were mixed into one cluster and showed a remarkable distinctness from all other populations. Individuals in the Erfoud5 population and Rich8 population form one large cluster with evidence of a weak separation of the two subpopulations visible based on the color coding. While individuals in the Erfoud6 population and Rich7 population were grouped into one group. These results were highly consistent with the results of UPGMA-based cluster analysis (Figure 3).

**Figure 2.** PCoA of the 280 individuals from 6 Moroccan alfalfa populations and 2 cultivars.**Figure 3.** UPGMA-based cluster analysis involving genetic distance between 6 Moroccan alfalfa populations and 2 cultivars.

We performed molecular phylogenetic analyses as a second method to analyze the general population structure that exists among genotypes of alfalfa. On the basis of Nei's (1978) [29] genetic distance, UPGMA-based cluster analysis was performed (Figure 3). The general tree topologies were similar to the PCoA results with two main groups (i.e cluster I and cluster II) in which the Demnate alfalfa populations (Demnate19 and Demnate21) were clearly separated from the Rich and Erfoud populations (Figure 3). Cluster I was divided into two subclusters (Ia, Ib). Subcluster Ia included the Demnate19 and Demnate21 populations. Subcluster Ib contained the Siriver cultivar. However, cluster II was further divided into three subclusters (IIa, IIb and IIc). Subcluster IIa contained Tamantit cultivar, whereas subcluster IIb grouped the Rich7 and Erfoud6 populations, and the sub cluster IIc grouped the Rich8 and Erfoud5 populations.

4. Discussion:

4.1. SSR polymorphism and genetic diversity:

To evaluate the informativeness and efficiency of SSR loci in analysis of genetic diversity as well as population differentiation assessments, the total number of alleles, H_E value, and PIC value for each SSR locus were estimated. All the SSR loci used in this study AFct32, MTLEC2A, AFca1, AFct11, B14B03, AFca11, and AFca16 were very effective for the analysis of genetic diversity in Moroccan alfalfa populations as revealed by the number of alleles (15, 16, 20, 28, 35, 12, 10) and heterozygosity (0.612, 0.399, 0.673, 0.788, 0.770, 0.642, 0.596), respectively (Table 3). The high number of alleles per locus per plant in alfalfa could be not only due to a very high level of heterozygosity but also because of the allogamous nature of cultivated alfalfa. These results confirm that markers with large number of alleles are informative for population studies. A low frequency of plants with either one allele or four different alleles compared to those with two or three alleles suggests that breeding programs were not carried out on Moroccan alfalfa populations. The seven loci analyzed in this work presented high PIC values (between 0.747 and 0.936), which were obviously related to the previously high number of alleles per locus. According to Botstein *et al.*, 1980 [27], who considered that $PIC > 0.5$ could indicate a highly polymorphic locus, we conclude that all SSR loci analyzed in the present study are highly polymorphic and informative group (average PIC of 0.848).

In this study, among the seven analyzed loci, the notable ones in terms of their ability in detecting genetic diversity are the AFct32 and the AFca16. On the one hand, AFct32 was the most effective locus for the analysis of genetic diversity in alfalfa studied populations, resulting in heterozygosity of 78.8%, and diversity among genotypes of 78.6%. According to Korkovelos A.E. *et al.*, (2008) [34], it is considered to be more effective primer that shows high values of heterozygosity and diversity index. On the other hand, AFca16 was considered as the less efficient locus in detecting genetic diversity in alfalfa. It showed the lowest number of alleles N_a (10), the highest inbreeding coefficient F_i (0.722), the highest number of null alleles r (0.269), and no correlation between estimated H_E (0.596) and observed H_o (0.165) number of heterozygous alleles. For this locus, we notice that there are fewer observed heterozygotes compared to the expected number of the latter. Having fewer heterozygotes at a locus suggests a departure from the expected Hardy-Weinberg equilibrium (HWE) proportions. Since only a single locus was concerned, this deviation is probably due to the occurrence of null alleles at this locus which is consistent with its low number of alleles per plant (1.17) and its highest number of null alleles r (0.269). The occurrence of null (non-amplifying) alleles can be the result of divergence in the sequences flanking the microsatellites that are undetectable on the unqualified DNA template. Thus, it causes heterozygotes to be mis-scored as homozygotes.

The genetic diversity of populations has an important impact on the evolution and adaptability of species. A lower level of genetic diversity will lead to a worse ability of species to adapt to the environment [35]. Abundant genetic resources provide alternative variation for breeding. A better understanding of genetic diversity may improve the utilization of related resources. The study worked on the genetic diversity and population structure of 6 cultivated Moroccan alfalfa populations and 2 introduced populations as accessed by SSR markers. The level of genetic diversity revealed by 280 genotypes in the study was as high as or even higher than that reported in similar studies on populations from other countries using SSR markers as reflected in average allele number per locus N_a (19.43), H_E (0.64), F_{ST} (0.263), H_s (0.650) and PIC (0.848). Falahati-Anbaran *et al.*, (2007) [36] studied the genetic structure of six cultivated alfalfa populations using eight SSR loci. their findings were 8.5 alleles per locus on average, F_{ST} of 0.024, H_s of 0.89 and PIC of 0.74. Bagavathianan *et al.*, (2010) [19] studied the genetic diversity of 22 alfalfa genotypes (populations, cultivars) using eight SSR markers and found 16.7 alleles per locus, H_E value of 0.77, H_s of 0.765 and F_{ST} of 0.0053. Quiang *et al.*, (2015) [37] evaluated the genetic diversity of 75 *M. sativa* sp. *sativa* accessions from a worldwide collection of alfalfa using 159 SSR primers and identified in average 8.96 alleles per locus, H_E value of 0.677, and PIC value of 0.638. Ilhan *et al.*, (2016) [38] investigated the diversity level of 70 accessions from the National Plant Germplasm Collection using 96 SSR markers and found 4.81 alleles per locus on average, H_E value of 0.35, H_s of 0.35, and PIC value of 0.28. He *et al.*, (2019) [15] studied genetic diversity of 242 alfalfa accessions using 102 SSR markers and noted a value of 4.62 as average allele number per locus, F_{ST} of 0.05, H_s of 0.514, and PIC value of 0.462. The different results registered among these studies could be either due to the differences between the origin of the populations being evaluated, the number of markers deployed, or the number of samples in the population. These results indicate that the Moroccan alfalfa populations studied show a higher genetic diversity.

High levels of genetic diversity in Moroccan alfalfa indicate that these populations are natural, and suggest that breeding programs were not carried out on these populations. Therefore, these alfalfa populations could be considered as valuable genetic material for alfalfa improvement programs. Alfalfa was initially introduced to Morocco around 100 years BC [39], especially in the Atlas mountain areas. Other more recent introductions were made in the 8th century and concerned the pre-Saharan oasis [39]. Since then, natural selection in diverse environments may have generated diverse germplasm as revealed in this study. It cannot be ignored that numerous local cultivars have been selected and formed by farmers. Indeed, traditional alfalfa cultivars such as Tafilalet, Rich, and Demnate have evolved according to a complex evolutionary process involving in particular natural and human selection and intra-specific gene flows, and this for several centuries [5]. These factors should be major reasons for the high genetic diversity in the Moroccan alfalfa populations used in this study even though the germplasm we used does not fully represent the existing diversity in Morocco.

In the present study, Erfoud5 population had the highest level of genetic diversity $A=9.571$, $G=16.429$, $H_s=0.723$, $H_E=0.705$, $I=0.151$, and $F_i=0.150$ (Table 4), whereas the Rich8 population had the lowest level of genetic diversity $A=6.571$, $G=12.857$, $H_s=0.575$, $H_E=0.558$, and $I=0.108$, and $F_i=0.200$. Alfalfa is an outcrossing species that suffers from inbreeding depression [40–41]. Inbreeding decreases the genetic diversity by increasing the frequency of homozygous genotypes and decreasing the frequency of heterozygous genotypes in the population [42]. It is evident, however, that inbreeding results in loss of heterozygosity and that F_i is related to this outcome [40]. According to Moss *et al.*, (2007) [43], when F_i value is <0.2 , it indicates low or moderate levels of inbreeding. Among the eight populations studied, all values of F_i computed were less than 20% indicating low levels of inbreeding within the studied alfalfa populations, except for the Erfoud6 and the Rich8 populations that showed a F_i value of 0.211 and 0.200 respectively. A F_i value of 0.200 indicates that approximately 20% of the loci are homozygous due to the inbreeding. This suggests that there is some loss of genetic diversity within the Rich8 population and Erfoud6 population, but is not extremely high.

4.2. Genetic differentiation and population structure:

Generally, the genetic variation in species with long life spans and outcrossing mainly occurs within populations, while the genetic variation in selfing species mainly occurs among populations [44]. In the present study, the AMOVA results showed that most of the genetic variance 56% was among individuals within populations, while 28% of the total variation occurred among populations (Table 5). This result indicates that the genetic variation in alfalfa mainly occurs within populations. Wright (1984) [45] provided clear rules for defining obvious genetic differentiation among populations ($F_{ST} < 0.05$, low; $0.05 \leq F_{ST} < 0.15$, medium; $F_{ST} \geq 0.15$, high). The study at hand revealed an F_{ST} value of 0.263 which means obvious genetic differentiation occurred among populations. Several other studies showed higher differentiation within than among alfalfa populations for neutral loci [12,19,36–38,46–47]. Allocation of most of the molecular genetics variance within populations is not surprising. Indeed, the allogamy of alfalfa, its autotetraploidy, its sexual propagation, and its local adaptation pressures result in large genetic variation at the intra-population levels. Human interference in seed process transfer eliminates effective eco-geological border to cause genetical discriminations [46]. All these factors promote the generation and maintenance of diverse genetic variants within the species. A low frequency of plants with either one allele or four different alleles, compared to those with two or three alleles and normal distribution of the total number of alleles within each plant suggest that breeding programs were not carried out on Moroccan alfalfa populations. Therefore, these alfalfa populations could be considered as valuable genetic material for alfalfa improvement programs. In addition, specific alleles were detected in Rich8 population and Tamantit population and could be useful for future studies like population identification and determination of the genetic purity of seeds. Private alleles generally support the potential to respond to selection or have evolutionary significance [48].

Population structure analysis enables us to understand genetic diversity in a given collection and identify the appropriate population for association mapping [37]. Cluster analysis indicated the ability and usefulness of SSR markers for studying the differentiation and relatedness among alfalfa populations. The results of UPGMA-based cluster analysis and PCoA showed that the Moroccan genotypes of alfalfa were clustered into two main groups that were significantly related to their geographical origins. One is composed of the Demnate alfalfa germplasm (high Atlas mountain), and the other is composed of the Rich and Erfoud alfalfa germplasm (desert valley of Ziz). The clear distinction between the populations of Demnate and those of the Ziz valley (Erfoud and Rich) suggests that the populations of these two regions do not have the same common ancestor. These results contribute to confirm the double genetic origin of the Moroccan material. Indeed, according to Birouk (1993) [5], he found that alfalfa in Morocco is divided into two gene pools depending on the origin. The first is made up of the populations of the High Atlasas well as the populations of Demnate. The second pool includes the Oasis areas and the desert valleys in addition to the populations of Rich and Erfoud. Cluster analysis also showed that individuals of Demnate populations were grouped with the individuals of Siriver cultivar, yet individuals of the Rich and Erfoud populations were clustered with the individuals of Tamantit cultivar. This distinction may be a consequence of the likely closed genetic background between them. Touil *et al.*, (2008) [49] displayed similar population structure using ISSR markers to analyze a genetic diversity of some Mediterranean populations of the cultivated alfalfa, one cluster grouped

populations of Demnate with Siriver, and another cluster grouped populations of Rich and Erfoud with Tamantit. Birouk (1993) [5] mentioned that populations of the high Atlas mountain were related to European origins and that those of Oasis areas and desert valleys were representative of the African type. The populations originated from Rich and Erfoud showed smallest genetic differentiation between one another (Table 5). A lack of distinct separation between Rich populations and Erfoud populations indicates that they share the same genetic history and that seeds were likely distributed extensively throughout valley of Ziz, ultimately leading to no or limited spatial structure. Indeed, as mentioned above, human interference in seed process transfer eliminates effective eco-geological border to cause genetical discriminations [46]. Another factor that might have influenced our results could be the low number of SSR markers used in this study, i.e. seven primers.

4. Conclusion:

In conclusion, the findings of this study underscore the utility of SSR markers for various applications in alfalfa research. These markers prove valuable for distinguishing between alfalfa populations, assessing genetic diversity both within and between populations, estimating genetic differentiation, and evaluating seed purity in alfalfa collections. The genetic data demonstrates a high level of diversity within Moroccan alfalfa, primarily concentrated within populations, which could serve as reservoirs for novel traits, although noticeable genetic differentiation exists among populations. Notably, private alleles were identified in the Rich8 and Tamantit populations, which could play a crucial role in population identification and ensuring the genetic purity of alfalfa seeds. Furthermore, these studied alfalfa populations represent a valuable genetic resource for potential plant improvement efforts, and steps should be taken to identify and preserve such resources. To strengthen our understanding, it is advisable to incorporate a larger number of populations from various geographical locations to confirm whether the clustering patterns observed here align with broader regional patterns. Additionally, the use of a wider range of SSR primers can enhance the precision of differentiation among closely related populations in future studies.

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